

Report
Of
Topic

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, Aradhya khapre do hereby declare that the internship report titled **“Topic”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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Roll No. 2410030543

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name: Aradhya khapre

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- Page no. of Sr. No. 1 to 6 along with a list of tables and list of figures will bear the Roman Numbering System (e.g. i, ii, iii).
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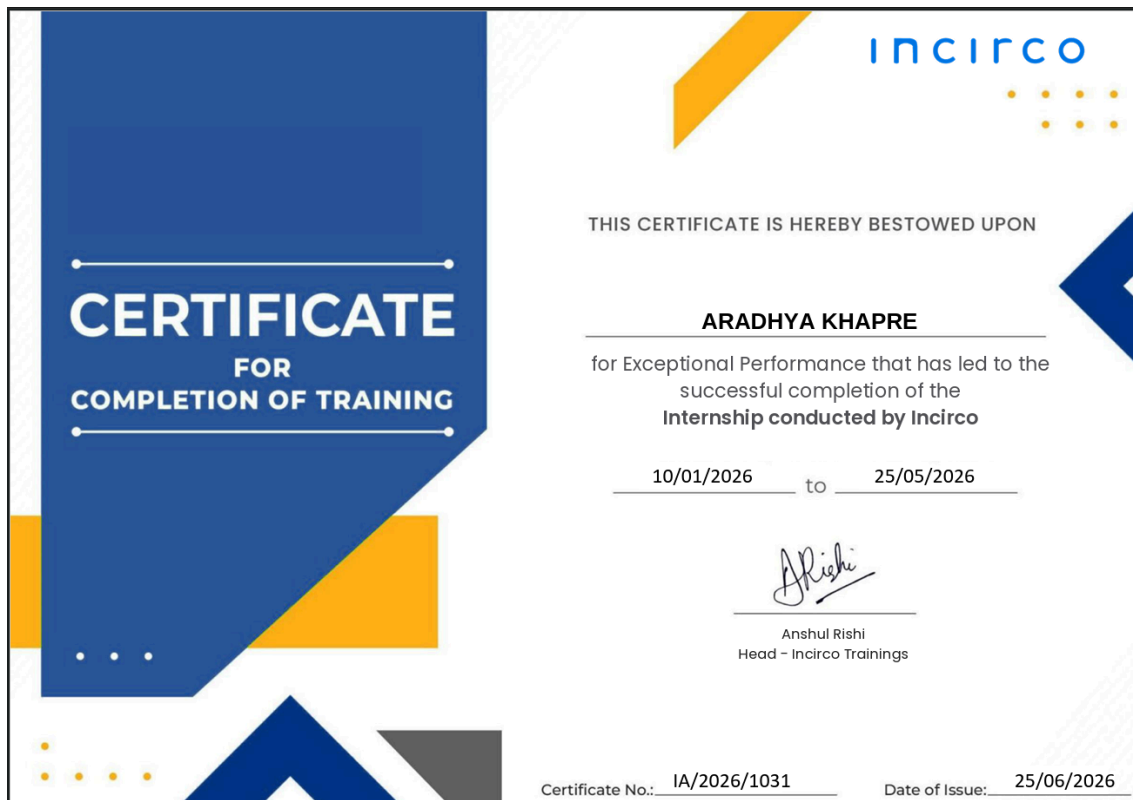
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Internship completion certificate



The certificate is a rectangular document with a white background. On the left side, there is a large blue vertical rectangle with a yellow triangle at the bottom right corner. Inside this blue rectangle, the text "CERTIFICATE FOR COMPLETION OF TRAINING" is written in white, bold, uppercase letters. To the right of this rectangle, the Incirco logo is displayed, consisting of the word "incirco" in blue lowercase letters and a yellow triangle to its right. Below the logo, the text "THIS CERTIFICATE IS HEREBY BESTOWED UPON" is written in a small, black, uppercase font. Underneath this, the name "ARADHYA KHAPRE" is written in a bold, black, uppercase font. Below the name, a line of text reads "for Exceptional Performance that has led to the successful completion of the Internship conducted by Incirco". Below this line, the dates "10/01/2026" and "25/05/2026" are written, separated by the word "to". Below the dates, there is a signature of Anshul Rishi. Below the signature, the name "Anshul Rishi" and the title "Head - Incirco Trainings" are written. At the bottom of the certificate, the text "Certificate No.: IA/2026/1031" and "Date of Issue: 25/06/2026" are written.

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Anshul Rishi
Head - Incirco Trainings

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4.1 Introduction

Enterprise Resource Planning (ERP) systems help organizations manage various business processes such as inventory, sales, finance, purchasing, and human resources through a single integrated platform. ERPNext, built on the Frappe Framework, is an open-source ERP solution that enables businesses to automate their operations efficiently.

During my internship at **Incirco Ventures LLP**, I gained practical exposure to ERPNext and the Frappe Framework by working in a professional enterprise software environment. The internship provided an opportunity to understand how ERP solutions are implemented and adapted to meet different business requirements while collaborating on real-world client projects.

The experience enhanced my understanding of enterprise application development and gave me insights into the complete ERP implementation lifecycle, including system configuration, customization, testing, documentation, and deployment support. It also strengthened my technical knowledge, analytical thinking, and problem-solving skills through hands-on learning.

4.2 organisation profile

Incirco Ventures LLP is a Delhi-based commerce operations consulting firm established in 2009. The organisation works with growing businesses in the e-commerce and retail space, focusing on the operational layer of commerce: order management, inventory control, marketplace integrations and ERP systems. Its stated positioning, "Operators, not advisors," reflects the firm's own experience of running consumer brands in apparel (Shellocks) and FMCG and home décor (isEver), which informs a hands-on, outcome-driven approach to client engagements rather than a purely advisory one.

The firm operates through three practice areas: Commerce Operations Advisory, Operations Performance, and ERP & Systems. Incirco is an Enterprise Solution Partner for both EasyEcom and ERPNext, and has published an EasyEcom and ERPNext integration application on the Frappe marketplace that is used by a growing base of Indian D2C and omnichannel businesses. In addition to consulting, the organisation runs Incirco Academy, a training vertical that shares its operational expertise with students and working professionals. Taken together, these activities position Incirco as a specialist firm that helps commerce businesses in India transition from improvised, manual operations to structured, system-led processes.

4.3 Problem statement

Organizations using ERP systems require accurate implementation, customization, and thorough testing to ensure that business processes function smoothly. As client requirements vary, ERP solutions must be validated for correctness, data consistency, and seamless integration with third-party platforms. Identifying software issues before deployment and ensuring reliable system performance are essential for minimizing operational errors.

The objective of this internship was to support these activities by assisting in ERPNext implementation, workflow validation, integration testing, documentation, and quality assurance, thereby contributing to the delivery of reliable and business-specific ERP solutions.

In addition to technical implementation, there was a need to ensure that the developed solutions aligned with actual business workflows and client expectations. This required careful verification of system functionality, preparation of supporting documentation, timely reporting of issues, and continuous validation of updates. Addressing these requirements was essential to improving the overall quality, reliability, and effectiveness of the ERP solutions delivered to clients.

4.4 Project Objectives

- Learn how an enterprise framework works.
- Understand the architecture and working of the Frappe Framework and ERPNext.
- Gain practical exposure to ERP implementation and customization.
- Learn enterprise business workflows such as Purchase Orders, Sales Orders, Goods Receipt Notes (GRN), Inventory, Payroll, and E-commerce Integration.
- Develop skills in software testing and quality assurance.
- Identify, document, and report software defects during application testing.
- Understand third-party integration between ERPNext and EasyEcom.
- Learn version control, issue tracking, and collaborative software development practices.
- Improve problem-solving, documentation, and analytical skills in an enterprise software environment.

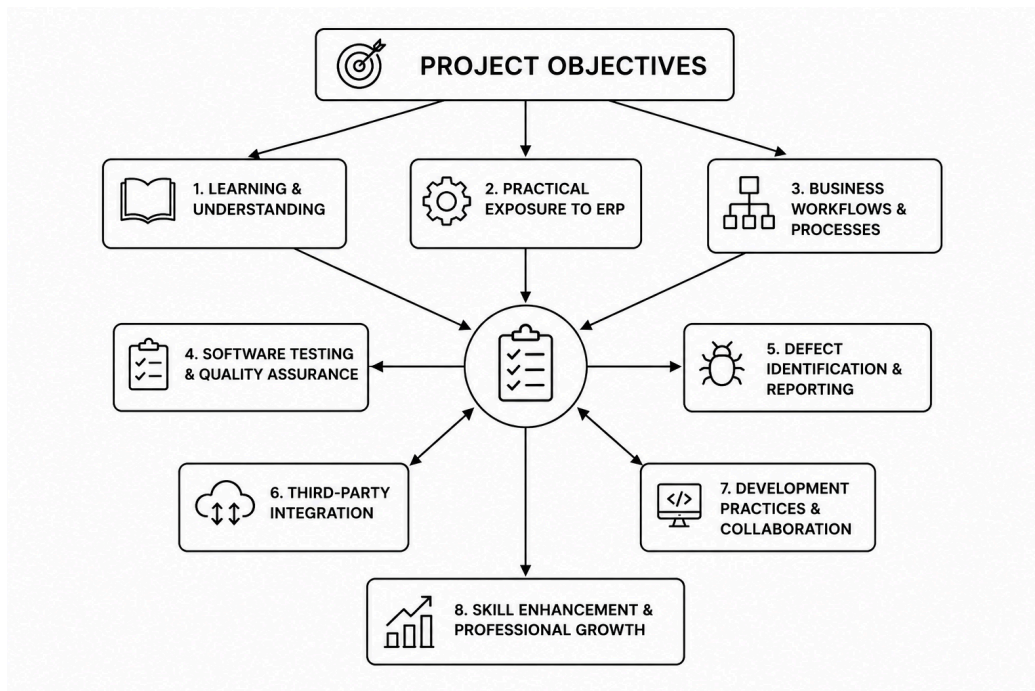


fig: 1.1 Flowchart

4.5 Scope of the project

The scope of this internship was to understand how enterprise applications are developed, customized, and maintained using the Frappe Framework and ERPNext. It provided practical exposure to real business environments by allowing me to work on different client requirements and understand how ERP solutions are implemented in organizations.

During the internship, I worked on various activities such as ERP customization, print format development, workflow testing, integration validation, documentation, and software testing. I also learned how different business processes are managed within an ERP system and how quality assurance plays an important role before deploying a solution.

Overall, the internship helped me build a strong foundation in enterprise application development and gave me practical experience that will be valuable for future projects and my professional career.

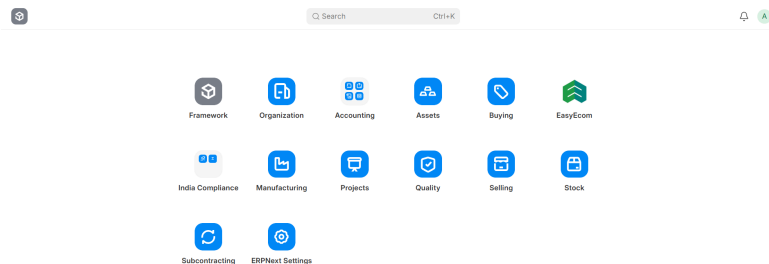


fig1.2 UI of ERPnext on frappe

```
alok@alokpc:~/frappe_bench$ bench start
13:57:43 system redis_cache.1 started (pid=2160)
13:57:43 system redis_queue.1 started (pid=2162)
13:57:43 system socketio.1 started (pid=2167)
13:57:43 system web.1 started (pid=2168)
13:57:43 system watch.1 started (pid=2172)
13:57:43 system schedule.1 started (pid=2177)
13:57:43 system worker.1 started (pid=2180)
13:57:43 redis_cache.1 2161:C 30 Aug 2026 13:57:43.122 # o080o080o080o Redis is starting o080o080o080o, modified=0, pid=2161, just started
13:57:43 redis_cache.1 2161:C 30 Aug 2026 13:57:43.122 # Redis version=7.0.15, bits=64, commit=08080808, modified=0, pid=2161, just started
13:57:43 redis_queue.1 2163:C 30 Aug 2026 13:57:43.122 # Configuration loaded
13:57:43 redis_queue.1 2163:C 30 Aug 2026 13:57:43.122 # Redis version=7.0.15, bits=64, commit=08080808, modified=0, pid=2163, just started
13:57:43 redis_queue.1 2163:C 30 Aug 2026 13:57:43.122 # Configuration loaded
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.122 # monotonic clock: POSIX clock_gettime
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.125 # Running mode=standalone, port=18002
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.125 # Server initialized
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.128 * Loading RDB produced by version 7.0.15
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.128 * RDB age 6580143 seconds
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.128 * RDB memory usage when loaded 1.23 Mb
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.130 * Done loading RDB, keys loaded: 11, keys expired: 3.
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.130 * DB loaded from disk: 0.003 seconds
13:57:43 redis_queue.1 2163:M 30 Aug 2026 13:57:43.130 * Ready to accept connections
13:57:43 socketio.1 Realtime service listening on: ws://0.0.0.0:9082
13:57:43 watch.1 yarn run v1.22.22
13:57:43 watch.1 error frappe-framework@ The engine "node" is incompatible with this module. Expected version ">=24". Got "18.20.8"
```

fig1.3 Running bench in WSL terminal

4.6 Technologies used

Technology/ Tools	Purpose
ERPNext	Used for understanding and working with enterprise business processes such as Sales, Purchase, Inventory, Payroll, and HR.
Frappe framework	Used to learn ERP application development, create custom DocTypes, configure roles and permissions, and perform customization.
Python	Used for backend customization and making changes to business logic where required.
JavaScript and JSON	Used for understanding and working with client-side scripting in the Frappe Framework.
HTML & CSS	Used for designing and customizing print formats and user interface elements.
EasyEcom Integration	Used for testing data synchronization and validating workflows between ERPNext and EasyEcom.
Git & GitHub	Used for accessing project repositories, reporting issues, and tracking bug fixes.
Windows Subsystem for Linux (WSL)	Used to set up the local Frappe development environment on Windows.
Visual Studio Code	Used as the primary code editor for development and customization.
Frappe Cloud	Used to deploy and test print formats and other ERPNext customizations.

4.7 System Architecture

The work carried out during the internship was based on the ERPNext platform, which is built using the Frappe Framework. Users interact with the system through a web browser, where requests are processed by ERPNext. The Frappe Framework handles the application logic, user authentication, workflow management, and business rules before communicating with the database.

The application stores all business-related information such as customer records, inventory details, sales orders, purchase orders, payroll data, and other enterprise information in the MariaDB database. For projects involving third-party services, ERPNext exchanges data with EasyEcom through APIs to ensure smooth synchronization of business information.

During the internship, my work mainly involved customization, workflow validation, testing, documentation, and integration verification within this architecture. Understanding this workflow helped me learn how different components of an enterprise application communicate to deliver reliable business solutions.

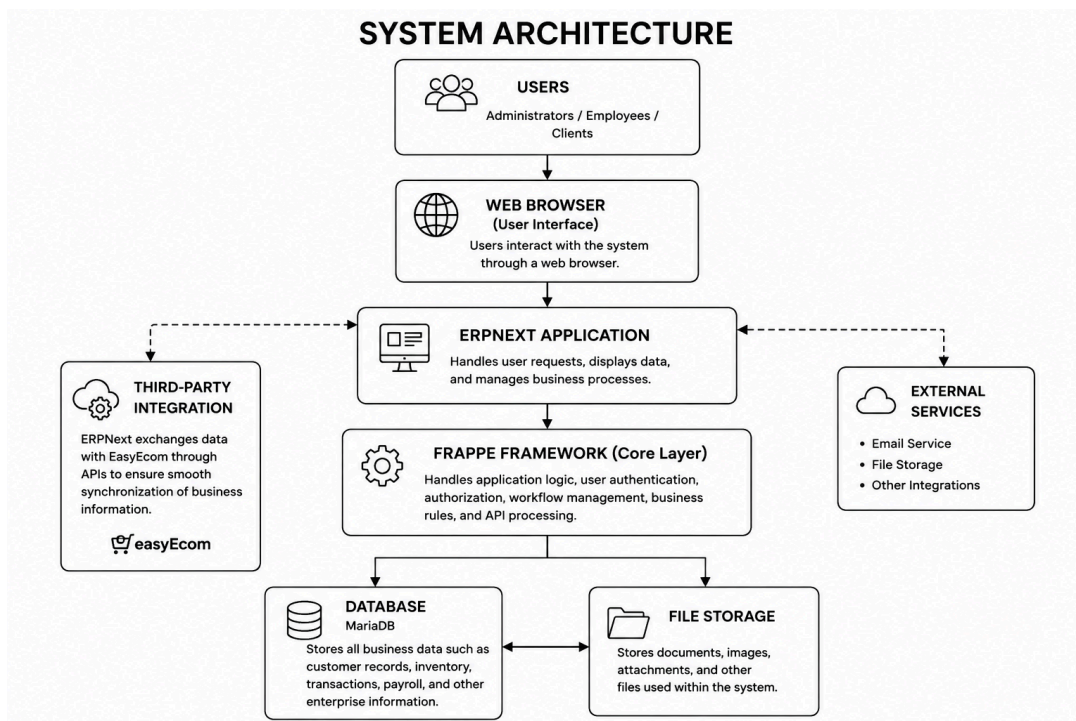


fig1.4 System architecture flowchart in details

4.8 Methodology

1. Learning the Frappe Framework

The internship began with understanding the basics of the Frappe Framework and ERPNext. I completed introductory courses from Frappe School and learned the fundamentals of enterprise applications, ERP modules, and the Frappe development framework.

2. Setting Up the Development Environment

After gaining the basic knowledge, I installed the Frappe Framework locally using Windows Subsystem for Linux (WSL). I created a local bench, site, and custom application, where I explored custom DocTypes, fields, roles, permissions, and dashboards to understand the framework practically.

3. ERPNext Customization and Implementation

Once familiar with the framework, I worked on tasks related to ERPNext customization, including print format development and understanding different business workflows. I also explored client requirements and learned how ERP solutions are configured according to business needs.

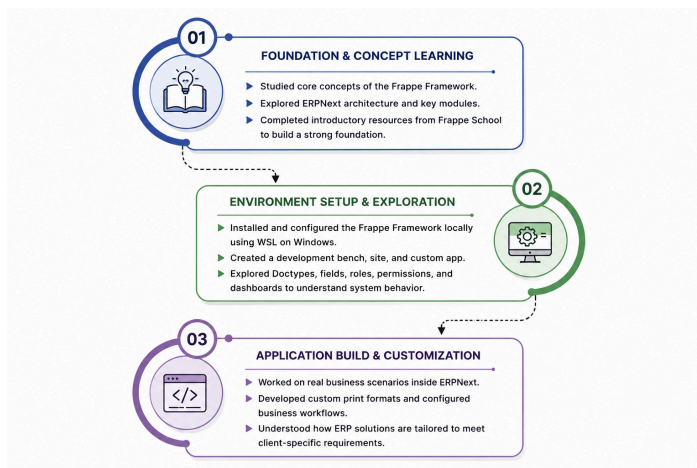


fig1.4 flowchart

4. Testing and Validation

A major part of the internship involved testing ERPNext workflows and validating the EasyEcom integration. This included verifying different business processes, cross-checking data for consistency, identifying issues, and ensuring that the implemented features worked as expected.

5. Documentation and Collaboration

Throughout the internship, I prepared documentation, attended project discussions, and collaborated with team members while working on live client projects. I also reported software issues, verified resolved bugs, and maintained updates through GitHub, which helped me understand the workflow followed in a professional software development environment.

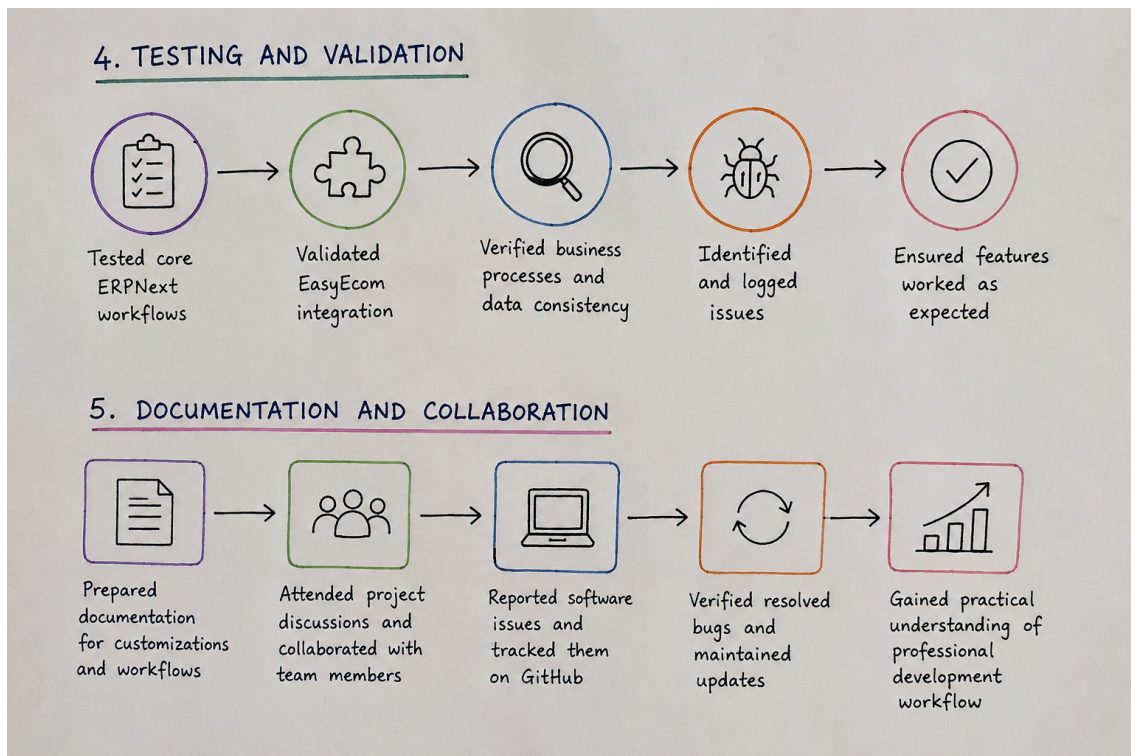


fig1.5 Content flow